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Optical Assurance Testing Specification GPON

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I ABBREVIATIONS, ACRONYMS, AND DEFINITIONS

ODN	**Optical distribution network**
ATP	**Acceptance test procedure**
PON	**Passive optical network**
iOLM	**Intelligent optical link mapper (EXFO proprietary)**
OLTS	**Optical loss test set**
OTDR	**Optical time-domain reflectometer**
FTS	**First Tier Split**
STS	**Second Tier Split**
ODF	**Optical distribution frame**
NAP	**Network access point**
ORL	**Optical Return Loss**
EPCM	**Engineering Procurement Contracts Management**

II RELEVANT AND APPLICABLE DOCUMENTS

Document No	**Description**
ITU-T G.984.1	GPON General Characteristics Specification
ITU-T G.984.3	GPON Transmission Convergence Layer Specification
IEC 61315	Calibration of fiber-optic power meters
ITU-T rec. g.650.3	Test methods for installed single-mode optical fibre cable links
ITU-T Rec. L.310	Optical fibre maintenance depends on topologies of access networks

1. Foreword

ISO 9000:2015(E)

In adherence to ISO 9000:2015(E) standards, an organization that prioritizes quality cultivates a culture that gives rise to behaviours, attitudes, activities, and processes aimed at delivering value by meeting the needs and expectations of customers and other relevant stakeholders.

The calibre of an organization's products and services is gauged by its capacity to satisfy customers and by its intentional and unintentional impact on pertinent stakeholders.

The quality of products and services encompasses not only their intended functionality and performance but also their perceived value and the advantages they offer to the customer.

2. Description

This document functions as a specification for optical testing methods within Fibertime PON networks. Additionally, it operates as a qualification method for newly acquired optical networks, aiming to establish confidence in the implementation model.

Considering the presence of several hand-off points, this document will also provide elaboration on both sectional and end-to-end testing procedures.

3. Scope

Optical assurance testing for Fibertime networks is restricted to the GPON network. The elements outlined below must adhere to the specifications outlined in this document to ensure compliance with the implemented performance standards. The stipulated requirements are as follows:

1. Receiver: Rx sensitivity
2. Transmitter: Tx Launch power, environmental effects
3. Passive optical connections: Element losses (splices, connectors, etc.)
4. Fiber types: Attenuation and environmental effects
5. Additional factors: Safety factors

4. Governance

EPCM Contractor:
The EPCM Contractor will be obligated to maintain records and provide access to databases.

Engineer:
The Engineer is expected to design in accordance with the standards and specifications stipulated by Fibertime Networks. Their responsibility encompasses ensuring that the contractor constructs based on an approved design that aligns with the standards and specifications of Fibertime Networks.

Engineer Representatives:
The Engineer Representatives are tasked with verifying that the implementation adheres to the designs provided by the Engineering contractor. This includes overseeing individual link constructions involving splitters, pre- and post-fiber qualifications. Additionally, they will ensure that the contractor consistently updates as-built records throughout the construction process, ultimately attaining completion and approval.

Contractor:
The Contractor is responsible for executing all plans in alignment with the standards and specifications of Fibertime Networks. They are also required to facilitate access to necessary qualification and construction progress data for all involved parties.

Fibertime Acceptance:
The internal Fibertime team holds the responsibility of accepting the final product into the production network.

5. Qualification and Assurance testing

5.1 Introduction

<u>Note that the cable runs will be referenced in Four sections.</u>

1. Primary Feeder
2. Secondary Feeder
3. Distribution
4. Drop

<u>Note that the splitters in this network are referenced in two tiers.</u>

1. First-tier split – between the primary and the secondary splitter
2. Second tier split – Between distribution and drop.

Note that a "link" is the section of fibre between OLT port and second tier splitter.

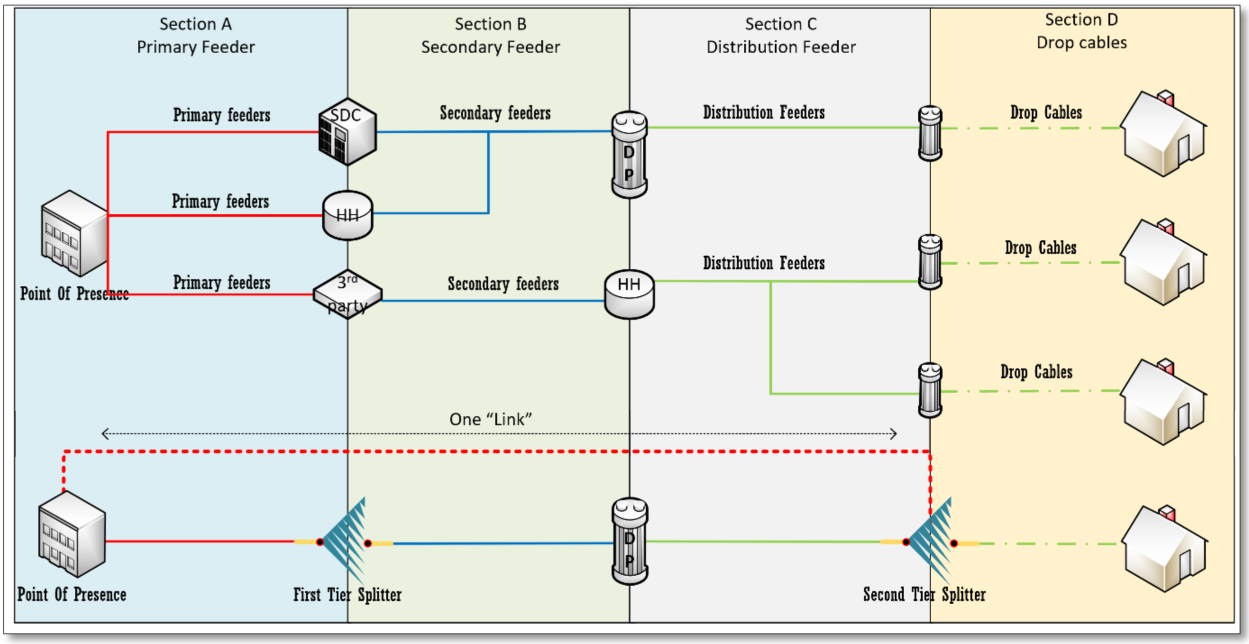

Figure 1: Network layout

A safety margin of **2 dB** is mandatory to factor in the aging of both active and passive equipment within the network. It is essential to emphasize that all equipment must be operated within the specifications provided by the vendor, in accordance with G.984 standards.

The following points below will delineate the specific aspects concerning the current Fibertime Networks network. However, it's worth noting that in the future, as Fibertime networks expand into different networking realms, additional testing methodologies might be necessary.

- **Acceptance Test Procedure (ATP) during Construction Phase**: This applies to cables and splitters.
- **ATP for Link Handover to Port**: Encompasses distribution up to the primary feeder's starting point.
- **ATP during Installation/Activation**: Pertains to ONT installation and provisioning.
- **ATP for Passive OLT Port in POP**: Addresses the preterm cable linking ODF and OLT.
- **Maintenance on Existing Full ODN**: Involves fault-finding test points in the operational network.

For an ATP to successfully meet the stipulated acceptance criteria, it's imperative that all criteria pass and are thoroughly documented.

5.2 Power Budget

The optical power budget is calculated to determine the minimum and maximum operating modes for a fiber network, relying on the power (dBm) introduced on each side and the sensitivity at the receiving end. The optical budget must undergo calculation to ensure that the network is constructed in accordance with standards across diverse attenuation distances and factors leading to loss.

In the context of optical power budgeting, attenuation is denoted in Decibels (dB), while Optical power is expressed in dBm. The table[Table 1] below provides insight into the Optical power budget.

The current operational framework of Fibertime permits a solitary optic classification, specifically C+.

Table 1: Power Budget

Power Budget: GPON Class C+				
	Maximum Receive Level	Minimum Receive Level	Maximum Power Budget	Minimum Power Budget
OLT (TX +5dBm) 1490nm	-12dBm	-32dBm	30dB	14dB
ONT (TX +2.5dBm) 1310nm	-8dBm	-26dBm	32dB	13dB

5.3 Link Loss Budget

The link loss budget should be calculated during the planning phase, taking into consideration the components listed below. The calculation should be performed using the calculator provided by Fibertime Networks. This calculation aims to determine insertion loss and ascertain whether a link passes or fails. A link, in this context, constitutes a GPON SFP transmitting to an ONT via a series of elements and then back to the GPON SFP.

For most scenarios, the 2nd tier split should mark the conclusion of the link loss calculation, as it will be closest to the drop. This distance is generally standardized to a maximum of 50 meters. In exceptional cases, the distribution length and drop length should be added to the overall distance for the sake of accurate calculation.

The link loss budget specific to each 2nd tier splitter needs to be included in the planning document. Additionally, these budgets should be securely stored in a database, which should also allow for queries from the Nokia easy ONT interface.

5.4 Maximum acceptable losses for elements

Table 2: Element Losses

Elements	Maximum acceptable losses (dB)
Head-End OLT and ONT	0.8 dB
Splitter 1:2	3.8 dB
Splitter 1:4	7.1 dB
Splitter 1:8	10.2 dB
Splitter 1:16	13.5 dB
Splitter 1:32	16.5 dB
Splitter 1:64	20.4 dB
Splice	Pass = < 0.23dB Warning = ≤ 0.25dB Fail = > 0.25dB
Mated connector Loss	0.5 dB
Loss 1310nm dB/km	0.4dB/km
Loss 1490nm dB/km	0.26dB/km
Loss 1550nm dB/km	0.3dB/km
Margin	2 dB

5.5 Connector reflectance

To ensure an acceptable Optical Return Loss (ORL), it is imperative to maintain reflectance within the specifications outlined below:

- Connectors with an Angled Physical Contact (APC) shall exhibit a maximum reflectance of -55dB.
- Connectors with a Physical Contact (PC) configuration shall not exceed a maximum reflectance of -40dB.

5.6 Optical Return Loss

A minimum ORL of 32dB is permissible within the network, and this value must be tested using either an Optical Time Domain Reflectometer (OTDR) or a well-calibrated Optical Loss Test Set (OLTS) tester.

5.7 EXPO Exchange and raw.trc files and reports

Upon the creation of jobs by the quality assurance team, all test results are required to be uploaded to EXFO exchange. Subsequently, these results will undergo standard processing procedures before being submitted to Fibertime Networks.

It is essential that all trace results are uploaded to a central database. Furthermore, comprehensive reports containing pass/fail determinations and other pertinent results, accompanied by photographic evidence, must be diligently compiled.

5.8 Cleaning optical connector

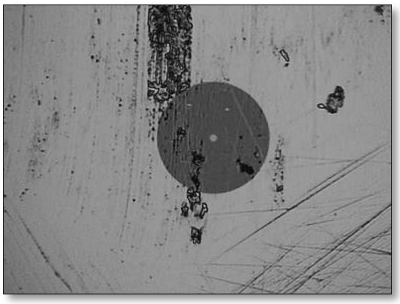

Figure 2: Dirty Scratched Connector

Maintaining clean optical connectors is of paramount importance to guarantee the seamless operation of the network as envisioned. Contaminated connectors stand as a significant concern in the realm of fiber optics, contributing to substantial connector loss, elevated reflectance levels, and even polluting transceivers. Remarkably, network issues often stem from unclean connectors, with reports from network operators suggesting that 15-50% of all network-related complications can be attributed to the presence of dirty connectors, ultimately leading to connection problems.

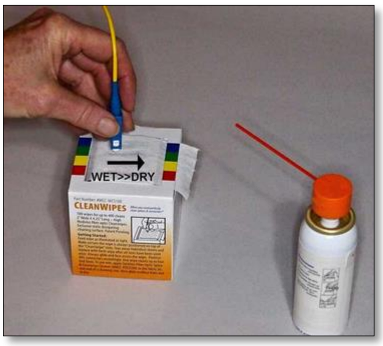

Figure 3:Wet to dry cleaning

In the realm of connector cleaning, the wet/dry method is typically favoured. This process involves two stages: wet cleaning, which serves to dislodge dirt and contaminants, and subsequent dry cleaning to eliminate them. Considerable research efforts have been dedicated to formulating cleaning solutions that excel in dirt removal, avoid generating static electricity, and prioritize safety.

Figure 4:Cleaning tools

In addition, the utilization of dry connector cleaners is considered secure. These cleaners involve treated lint-free tapes housed within cassettes, boxes, or compact handheld tools (probes). These tools facilitate cleaning connectors within mating adapters, even those situated at the far end. Certain probes are engineered to clean within mating adapters, although it is often recommended to utilize a specialized swab for this purpose. Notably, specialized cleaners have been designed to counteract the challenge of static electricity, which can attract dirt to connectors that have just been cleaned.

6. *Types of testing equipment*

6.1 OTDR and iOLM

Figure 5:OTDR/iOLM

An OTDR unit with valid calibration certificate will be required, it must be capable of testing 1310nm and 1550nm and MUST be iOLM enabled.

6.2 OLTS Test set

Figure 6:OLTS test set

To achieve precise Optical Return Loss (ORL) and Insertion Loss (IL) measurements on the link, an Optical Loss Test Set (OLTS) is essential. This instrument enables accurate measurements using the two necessary wavelengths: 1310nm and 1550nm, and it incorporates a loopback function. Additionally, this module can be seamlessly integrated into a conventional Optical Time Domain Reflectometer (OTDR) chassis.

It's imperative that the EXFO equipment used is calibrated and certified by an authorized calibration laboratory. The provision of a calibration certificate substantiates the accuracy and reliability of the equipment.

6.3 PON power meter

Figure 7:PMO device

A PON power meter with in-line function will be required with a calibration certificate.

The PON power meter plays a crucial role in fault identification within the operational network, particularly when an Optical Network Terminal (ONT) fails to auto-provision. This tool is equally indispensable for ongoing network maintenance activities.

For these purposes, a PON power meter equipped with an in-line function is essential. It is pivotal that this meter is accompanied by a calibration certificate, confirming its accuracy and reliability.

6.4 Divot Testing – Bare Fiber Adapter (Tester)

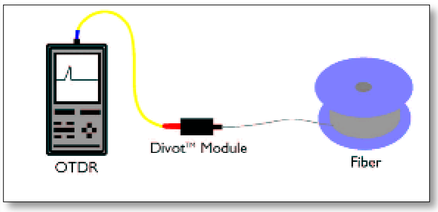

Figure 8: Divot Testing Device

A Divot tester serves as a device equipped with mating gel enclosed within a module. Its design is tailored to enable the connection of a bare, cleaved fiber to a connector, facilitating continuity testing. It's worth noting that general reflection levels are somewhat inferior compared to testing using the connectorized approach. A cartridge integrated into this device can typically endure a minimum of 500 insertions, and its replacement is a straightforward process.

Periodically, maintenance may involve cleaning both the ferrule on the patch cable and the ferrule within the Divot® Module.

6.5 EXFO Calibrations

Fibertime mandates that all EXFO testing equipment undergo annual calibration at an EXFO-approved centre, in accordance with the manufacturer's guidelines. Calibration dates will be visible within the EXFO exchange software, and the corresponding calibration certificates must be uploaded annually by each contractor.

7. Testing procedures

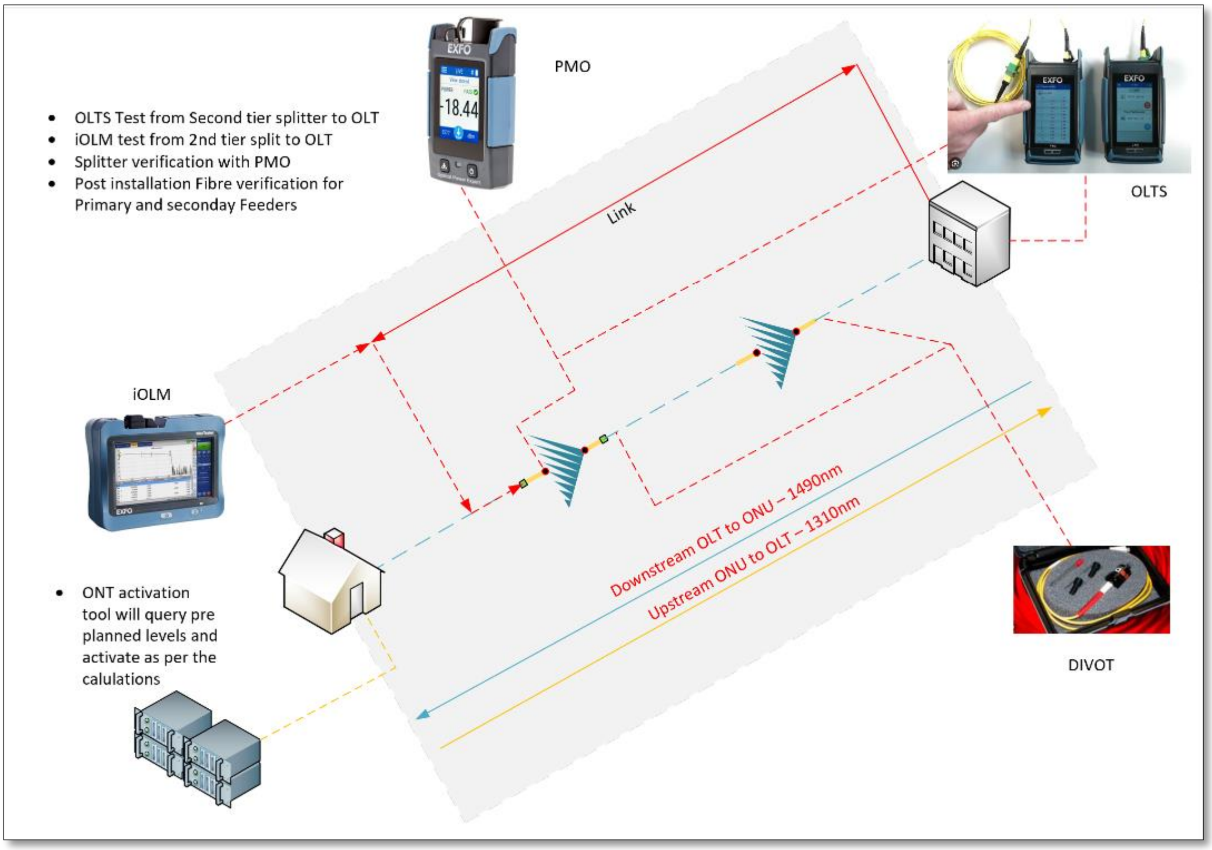

Figure 9: Optical test layout

This section outlines the distinct segments of the network tested during each phase. These procedures are relevant for networks regardless of whether they have active equipment in operation or are devoid of any equipment.

7.1 OLTS Testing

During the handover of a link, it's imperative to conduct an Optical Loss Test Set (OLTS) test using a bi-directional approach. This test will span from the second-tier splitter up to the ODF primary feeder termination. The requisite wavelengths for this test are 1310nm and 1550nm. The loss values on the devices must be expressed in decibels (dB).

To ensure accurate results, both testing devices need to be referenced to zero before the testing process begins. Additionally, adhering to clear link indication naming is crucial for effective file naming. For instance, "A-B" and "B-A" should be designated on each device. In terms of device location identification, on Device A, the ODF should be noted using the Cable Identifier_Fibre Number_POP Name_2nd Tier Splitter Name. Conversely, on Device B, the Pole Identifier_2nd Tier Splitter_Splitter Zone Name should be employed.

The first output on the 2nd tier splitter will be utilized for the OLTS test. In scenarios where an acquired network displays variations in the setup, where first tier and second-tier splitters are positioned differently, a logical approach should be adopted.

Conducting OLTS testing verifies the optical network's compliance with active component requirements. It encompasses testing from the point closest to the OLT port (primary feeder termination) to the point closest to the customer (including inside unit TP). This meticulous testing guarantees the successful implementation of an optical network.

7.2 iOLM/OTDR Testing.

- *Link Testing*

  In addition to the OLTS test, handing over a link will necessitate an iOLM (intelligent Optical Loss Measurement) test performed through the second-tier splitter up to the primary feeder termination. For this test, a launch coil of 100 meters is necessary to facilitate testing through the second-tier splitter.

  For this test, the following file naming convention will be applied: **Test Type_Pole Identifier_2nd Tier Splitter_Splitter Zone.**

  In cases where an iOLM test is conducted, it's vital to include all raw traces along with the test results for submission.

- *Pre- and post-installation testing*

  For both primary feeder and secondary feeder cables, a fiber integrity test is mandatory at 30 seconds per fibre after installation. This test can be effectively conducted using an OTDR test in the field. The results should be provided in RAW format and accompanied by the OTDR report.

  For **primary feeder testing,** the procedure involves initiating the test from the termination point within the Point of Presence (POP) while utilizing a 100m launch coil. The goal is to assess the quality of splicing within the Optical Distribution Frame (ODF) and to verify the integrity of all fibers within the cable. This test will occur post-termination on the ODF side and pre-termination on the manhole breakout side.

  In contrast, **secondary feeder testing** entails utilizing a "Divot" from the manhole breakout to the termination point on the distribution cable. This test will be carried out pre-termination after installation.

  For file submissions, the following naming convention is acceptable: Cable **Identifier Fibre Number A Location-B Location.**

7.3 Power meter Testing

Power meter tests are integral to ascertain splitter integrity, essentially confirming that all output and input legs operate within defined parameters post-installation. This evaluation will occur on the connectorized side of the second tier split during the pre-commissioning phase, following the installation.

For this test to succeed, all measurements must align with the calculated link loss budget furnished during the planning phase. In scenarios where POP equipment isn't activated for appropriate power levels, the contractor has the option to use a light source for injecting PON downstream wavelengths at 1490nm. This light will be measured both before the splitter input and at the same splitter output.

If launching is performed with live equipment, a C+ optical launch of ±5dBm is required. Alternatively, launching with a non-transmission device permits a 0dBm launch while applying a 5dB budget.

File naming for these tests adheres to the format: **Test Type Pole Identifier_2nd Tier Splitter Splitter Zone.**

In the context of POP OLT port testing, an OLT port power meter result will be mandated on the ODF side, post sign-off. This measurement will determine the power output (dBm) of the SFP and validate that all ports and SFPs adhere to manufacturers' specifications post activation.

File naming for this test follows this pattern: **OLT Identifier_OLT Port_Tray Identifier_Fibre Number.**

7.4 ATP File

For every zone that is being handed over, an Acceptance Test Procedure (ATP) must be developed and adhered to.

The ATP should encompass a comprehensive compilation of reports derived from FastReporter files. It must also integrate all power meter results, along with accompanying reference photos. Furthermore, the ATP must contain a comprehensive and signed table that encompasses all Power Meter (PM) and Light Source results, and this should include a calculated link loss budget as well.

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